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NATs



- **NAT (Network Address Translation)** is a way for devices at home (like your phone, laptop, smart TV) to share one single internet address.
- **Why do we need NAT?**
- **IP addresses are limited:** There aren't enough internet addresses for every device in the world. NAT helps by letting many devices share one address.
- **Extra security:** It also blocks random people from reaching your devices unless you ask for something first (like visiting a website).

How it works:

- Inside your home (your LAN — Local Area Network), all your devices have private, local addresses (like 192.168.1.5, 192.168.1.6, etc.).
- When a device wants to go online, the router (your Linksys, D-Link, or Netgear box) changes the local address to your **one public address** — the one the world sees.
- The router keeps track of who asked for what, so when a website sends data back, it knows which device to send it to.

NAT Types:

- NAT often uses private addresses like **10.x.x.x** or **192.168.x.x** inside your home.
- Outside, you have **one public IP** shared by all devices.

Two main kinds:

- **Address Translation** — just change the IP address.
- **Address and Port Translation (NAPT)** — change **IP + port number** (very common today!).
- **NAPT = most common NAT today** (used in almost all home routers).

Extra detail:

- NAT changes info in the **IP header** (the address) and **TCP/UDP header** (the port).

Will NAT work with IPSec?

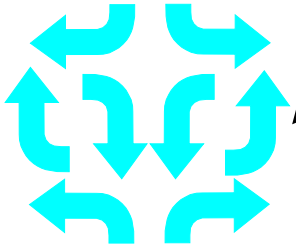
- **Not easily.**
- IPSec encrypts the data, so NAT can't read or change it properly.
- Special tricks (like NAT-T) are needed to make IPSec work with NAT.

Example

- Imagine you have a laptop at home with the private IP address **192.168.1.10**. When you open a website like Google, your router uses **NAT** to hide your private address. It changes your laptop's IP to your public IP, like **203.0.113.5**, and also changes your port number to something random, like **54321** — this is called **NAPT**. Now, when your request goes to Google, it looks like it's coming from **203.0.113.5:54321**. When Google sends a reply, your router knows how to match it back to your laptop.

NAT Example

IAP's Point of Presence



Messages sent between host B
to another host on the Internet
Host B original source socket:

192.168.0.101 port 1341

Host B translated socket:

68.40.162.3 port 5280

Router with NAT
External IP: 68.40.162.3
Internal IP: 192.168.0.0

Router assigns internal
IPs to hosts on LAN :

A: 192.168.0.100

B: 192.168.0.101

C: 192.168.0.102

- In IPSec, we use a **Security Association (SA)** to provide either **authentication** (AH) or **encryption** (ESP). To do both, we combine SAs into a **Security Association Bundle**.
- SAs can be combined by **transport adjacency** (one after another) or **iterated tunneling** (wrapping in multiple tunnels). They may end at the same or different devices.
- The best order is to **encrypt first, then authenticate**, so both the data and encryption are protected.

Security Association Bundle

- A **Security Association Bundle** means more than one **SA** is used on the same packet.
For example, **ESP** encrypts the packet but **does not protect** the new IP header.
To fix this, we **use two SAs**:
 - First SA: apply **ESP** (for encryption only).
 - Second SA: apply **AH** (to add authentication, including the IP header).
 - This way, the packet is both **encrypted and authenticated** properly!

Outbound Packet Processing...

- When sending a packet **outbound**, an **Integrity Check Value (ICV)** is created to check if the data is tampered with. The ICV includes everything in the **ESP packet** except for the **authentication data** itself. To make sure the ICV algorithm works properly, the packet is padded with extra '0's between the **next header** and the **authentication data**. This padding ensures the data fits the required size for the ICV calculation.

For **inbound packet processing**, the **sequence number** is checked first. This is part of **anti-replay protection**, which is used only if **authentication** is selected. The router looks at the **sequence number** to see if the packet is a **duplicate** — and if it is, it's **rejected**.

The **anti-replay feature** is optional and the info is stored in the **SA entry**. It uses a **32-bit sequence number** for outgoing IPSec packets. There's also a **32-bit anti-replay window** with a **bit-map** to detect and reject **replayed packets**.

Key management

Key management handles key generation and distribution. Typically, you need **2 pairs of keys**: one for each direction (in and out) for **AH** and **ESP**.

- **Manual key management**: A sysadmin manually configures keys for each system.
- **Automated key management**: An automated system creates keys on demand for **large systems** with many SAs.

Question?